

APPLIED MACHINE LEARNING AND NEUROSymbolic AI

Engagement Letter & Statement of Work

Project NESY-AML-06 — Multi-Hop Question Answering over Knowledge Graphs

Two-Track Assignment: Applied ML (Track A) & Neurosymbolic AI (Track B)

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001

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Assignment type:

☒ Individual

☐ Team of 2

Engaging Party
("Client")

Name:

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Role / Title:

CEO

Institution & Department:

Cybersecurity Company

Assigned Professional(s)
("Contractor")

Name:

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Second team member (if any):

Supervising Professor

Name:

Track(s) assigned

☐ Track A only

☐ Track B only

☒ Both tracks (recommended)

Dear Assigned Professional,

We are pleased to confirm your engagement to carry out the analytical and technical work described in this Statement of Work (the "Engagement"), issued under the course *Applied Machine Learning and Neurosymbolic AI*. This letter sets out the purpose, scope, deliverables, timeline, and evaluation criteria governing the Engagement, and constitutes the agreed basis on which the work will be assessed.

1 Purpose and Scope of the Engagement

The purpose of this Engagement is to produce two independent, professionally documented solutions to the same multi-hop knowledge-graph question answering problem: one grounded in the classical, non-deep Path Ranking Algorithm (Track A), and one grounded in a hand-specified symbolic semantic parser executed by a purely symbolic graph-traversal engine (Track B). The two tracks are to be conceived, implemented, and evaluated **independently of one another** on the same benchmark; Track B must not import, fine-tune, or otherwise reuse any trained ranker, learned path features, or code artifact produced under Track A. The Engagement concludes with a comparative analysis contrasting the two independently produced solutions.

2 Background

Many real questions cannot be answered from a single knowledge-graph fact: answering “Who directed the films starring the actor who played character X?” requires chaining several relational hops from a starting (“topic”) entity to a final answer entity. Multi-hop knowledge-graph question answering (KGQA) is therefore a natural setting in which to contrast a *statistical, path-feature-based* approach to relational reasoning with a *fully symbolic, grammar-and-traversal-based* approach, since both must solve the same underlying problem — finding a correct relational path from topic entity to answer — using fundamentally different machinery.

3 Problem Statement

Given a natural-language question, a knowledge graph $\mathcal{G} = \{(h, r, t)\}$, and a topic entity mentioned in the question, the task is to return the correct set of answer entities reachable from the topic entity via a k -hop relational path implied by the question ($k \in \{1, 2, 3\}$). Systems are evaluated separately on each hop-count subset, since difficulty increases sharply with the number of hops.

4 Data Sources and Description

Primary dataset

MetaQA, a large-scale multi-hop KGQA benchmark in the movie domain (Zhang et al., 2018), built on a WikiMovies-derived knowledge base. The knowledge graph contains **43,234** entities (movies, actors, directors, writers, genres, languages, release years, and more), **9** base relation types, and **134,741** triples. Entity degree is heavily long-tailed: the median entity has only 4 triples, while a small number of hub entities (highly prolific actors, for instance) reach up to roughly **4,431** triples — directly responsible for the path-explosion challenge described below.

MetaQA provides **407,513** natural-language questions in total, split by hop count, each with its own train/dev/test partition:

	Train	Dev	Test
1-hop	96,106	9,992	9,947
2-hop	118,980	14,872	14,872
3-hop	114,196	14,274	14,274

The Engagement requires use of the *vanilla* text variant (clean, template-generated natural-language questions), as opposed to the noisier NTM-text or audio variants, so that both tracks’ comparison isolates multi-hop relational reasoning rather than speech- or paraphrase-robustness. Use of the noisy variants is an optional bonus extension.

Optional extension dataset

WebQuestionsSP (WebQSP): 4,737 questions requiring up to 2-hop reasoning over Freebase, with gold SPARQL queries (Yih et al., 2016). Suitable as a bonus cross-domain generalization test for either track once the MetaQA pipeline is validated.

Known data challenges

- **Combinatorial path explosion.** High-degree hub entities (up to $\sim 4,431$ triples) cause the number of candidate relational paths to explode at 2 and especially 3 hops, directly threatening the tractability of Track A’s path-ranking approach.
- **Topic-entity linking.** Reliably identifying the correct starting entity mentioned in the question text is itself a non-trivial sub-problem shared by, but to be solved independently within, both tracks.
- **Relation ambiguity.** Only 9 base relation types exist, but multiple distinct relational paths can plausibly connect the same pair of entity types, requiring careful feature or rule design to disambiguate.
- **Sharp difficulty gradient.** Accuracy on published KGQA systems typically drops substantially from 1-hop to 3-hop questions, which is itself one of the central phenomena this Engagement asks both tracks to characterize.

5 Scope of Work — Track A (Applied ML, Pure/Classical)

Objective

Implement the classical **Path Ranking Algorithm (PRA)** to answer multi-hop questions via random-walk path features and a classical statistical ranker.

Required approach

- Implement a simple, deterministic topic-entity linker (e.g., exact or fuzzy string matching against the KG’s entity dictionary).
- From each topic entity, enumerate bounded-length relational paths (capped at the question’s hop count) and compute PRA-style random-walk path features, i.e., the probability of reaching each candidate entity via each distinct relation-path type.
- Design and document an explicit **path-pruning strategy** (e.g., frequency-based pruning, maximum branching factor, path-length capping) to keep computation tractable at 2 and 3 hops — this is the central engineering challenge of the track and must be justified, not merely defaulted.
- Train a Logistic Regression ranker over the extracted path features to score candidate answer entities, per hop-count subset.
- Evaluate with Hits@1 and Mean Reciprocal Rank (MRR), reported separately for 1-hop, 2-hop, and 3-hop questions.

Explicitly out of scope for Track A

Any knowledge-graph-embedding method (TransE, ComplEx, RotatE, DistMult) and any neural multi-hop QA architecture (Key-Value Memory Networks, GraftNet, PullNet, EmbedKGQA, or

any Transformer-based reader) are excluded. These are standard neural baselines in the published KGQA literature and may be cited for context, but Track A’s purpose is to characterize what the classical, feature-based PRA approach — with no learned embeddings and no neural components — can achieve on its own.

Track A deliverables

The entity-linking module; the PRA feature-extraction pipeline with a documented and justified path-pruning strategy; the trained Logistic Regression ranker(s); a Hits@1/MRR evaluation report broken down by hop count; a written analysis of how and why accuracy degrades as hop count increases.

6 Scope of Work — Track B (Neurosymbolic AI, Independent)

Objective

Build a fully symbolic question-answering pipeline: a hand-specified semantic parser mapping questions to an executable multi-hop graph-traversal query, developed independently of Track A.

Required approach

- Because MetaQA’s vanilla questions are template-generated, design and document a small hand-specified lexicon/grammar mapping trigger phrases and syntactic templates in the question to a formal relation-sequence query (i.e., an ordered chain of relation types to traverse from the topic entity).
- Implement your own topic-entity linker within this track (it must be a separate, independent implementation from Track A’s, even if similar in design, so that the two tracks remain self-contained).
- Execute the resulting query as **pure symbolic graph traversal** over the KG triples — e.g., breadth-first multi-hop traversal implemented directly, or via a Prolog/Datalog-style query engine (`pyDatalog` or an Answer Set Programming encoding in `clingo`). No statistical scoring, ranking model, or learned weight of any kind may enter this track.

Track B deliverables

The documented lexicon/grammar, including its coverage over the question templates observed in the training data; the symbolic query-execution engine; the same Hits@1/MRR evaluation protocol as Track A, broken down by hop count, for a direct comparison; an analysis of parser coverage (the fraction of questions the grammar correctly parses into an executable query) and of the main failure modes (unseen phrasings, entity-linking ambiguity, template gaps).

7 Comparative Analysis Requirement

The final report must directly contrast Track A and Track B along at least the following axes:

- Accuracy versus number of hops: does a correct symbolic parse in Track B degrade less across hop count than Track A’s random-walk-based ranking, whose path features grow noisier as path length increases?
- Explainability of the answer: an explicit, human-readable traversed relation chain (Track B) versus a weighted combination of path features (Track A).

- Robustness to unseen question phrasings: statistical generalization (Track A) versus grammar coverage (Track B), discussed qualitatively with concrete failure examples from each track.
- How topic-entity-linking errors propagate differently through a statistical ranker versus a deterministic symbolic traversal.

8 Deliverables Summary

1. Technical report (Track A and Track B sections, plus the comparative analysis), in PDF.
2. Two independent, well-documented codebases (or a single repository with clearly separated Track A / Track B directories and no shared trained artifacts).
3. Reproducibility package: environment specification, instructions to regenerate all reported numbers from raw data.
4. Short oral presentation (slides) summarizing both tracks and the comparison.

9 Timeline and Milestones

Week	Track A milestone	Track B milestone
1–2	Data acquisition, exploratory analysis of the KG and question templates; entity-linker prototype.	Same shared data understanding; cataloguing of question templates for grammar design; independent entity-linker prototype.
3–4	PRA path enumeration and feature extraction on 1-hop questions; pruning-strategy design.	Lexicon/grammar design for 1-hop and 2-hop templates; traversal-engine prototype.
5	Extension to 2-hop and 3-hop questions; Logistic Regression ranker training and tuning.	Grammar extension to 3-hop templates; coverage analysis and refinement.
6	Full Hits@1/MRR evaluation per hop count; error analysis.	Full Hits@1/MRR evaluation per hop count under identical protocol; coverage and failure-mode analysis.
7	Joint comparative analysis write-up (both tracks contribute results).	
8	Report drafting and code cleanup / reproducibility package.	
9	Instructor review checkpoint; revisions.	
10	Final submission and presentation.	

10 Evaluation Criteria

Criterion	Weight	Description
Track A technical execution	30%	Soundness of the PRA implementation, path-pruning strategy, and ranker tuning.
Track B technical execution	30%	Quality and coverage of the grammar/lexicon, correctness of the symbolic traversal engine.
Comparative analysis	20%	Depth and evidence base of the Track A vs. Track B discussion, especially the hop-count degradation analysis.
Report clarity & reproducibility	15%	Writing quality, documentation, ability to reproduce results from the package.
Presentation	5%	Clarity and quality of the oral summary.